

Campus Thrift Store – Project Report

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Problem Statement:

Students often have items such as clothes, accessories, shoes, electronics, and books that they no longer need but are still in good condition. At the same time, other students look for affordable second-hand items within their campus community. However, there is no simple, centralized, and student-friendly platform to facilitate this exchange.

The Campus Thrift Store web application addresses this problem by providing a mini online marketplace specifically designed for students. The platform allows users to browse available second-hand items and enables students to list their own items for sale through an interactive form. The system focuses on simplicity, usability, and sustainability by encouraging reuse and reducing waste. The project is implemented using only frontend technologies (HTML, CSS, and JavaScript) and demonstrates practical web development concepts such as DOM manipulation, form validation, responsive design, and interactive user interfaces.

Web Design & UI/UX Approach:

The user interface of the Campus Thrift Store was designed with a clean, modern, and minimal aesthetic to ensure ease of use and visual clarity. A neutral color palette with high contrast was used to keep the focus on product images and content. The layout is consistent across all pages, featuring a fixed navigation bar at the top for easy access to different sections of the website.

Responsive design techniques were implemented using CSS Grid and Flexbox to ensure the website adapts smoothly to different screen sizes, including mobile devices. Large hero sections with overlay effects are used to create visual impact, while hover animations on product cards and buttons enhance user interaction. Typography is kept bold and readable, especially for headings and prices, to improve usability. Overall, the design prioritizes clarity, accessibility, and an engaging shopping experience for students.

Key Pages and Functions:

Home Page (index.html):

The Home page serves as the landing page of the website. It introduces the Campus Thrift Store with a hero banner and highlights featured and most-loved products. JavaScript dynamically loads selected items into the featured sections using DOM manipulation. The page also includes category navigation and a call-to-action section encouraging users to start selling their items.

Marketplace Page (items.html):

The Marketplace page displays all available items in a structured grid layout. Items are dynamically rendered using JavaScript by combining predefined products and user-added items stored in the browser's localStorage. Each product card displays the item image, name, price, category, description, and a "Buy Now" button, which redirects users to the contact page.

Sell Page (sell.html):

The Sell page contains the main functional form of the application. Users can enter item details such as name, category, price, description, and upload an image. JavaScript validates all inputs and uses the FileReader API to convert uploaded images into Base64 format. Successfully submitted items are stored in localStorage and become visible in the Marketplace page.

Contact Page (contact.html):

The Contact page allows users to send queries or express interest in buying an item. It includes a simple form with client-side validation and displays a confirmation alert upon successful submission.

Special Features:

DOM Manipulation:

JavaScript is used extensively to dynamically create and display product cards on both the Home and Marketplace pages. The `createCard()` function generates HTML elements at runtime, allowing items to be added without refreshing the page. Featured and marketplace items are loaded dynamically based on page context.

Form Validation:

Client-side validation ensures that all required fields in the Sell and Contact forms are completed correctly. Empty inputs, missing images, or invalid data trigger alert messages, preventing incorrect submissions and improving data quality.

Local Storage Usage:

The application uses the browser's `localStorage` to store items added by users through the Sell page. This allows newly listed products to be displayed instantly on the Marketplace page using JavaScript and DOM manipulation, without requiring a page reload. The use of `localStorage` demonstrates client-side data persistence within a frontend-only web application and supports interactive functionality during user sessions. Since the project is implemented using only HTML, CSS, and JavaScript, `localStorage` serves as a practical solution to simulate item storage without backend integration. However, `localStorage` is browser- and device-specific, meaning that items added on one computer are only visible on that same system. To ensure consistency during demonstration and evaluation, the application also includes a predefined set of sample items embedded directly in the JavaScript code. In a real-world deployment, a backend server and database would be required to enable shared data access across multiple users and devices.

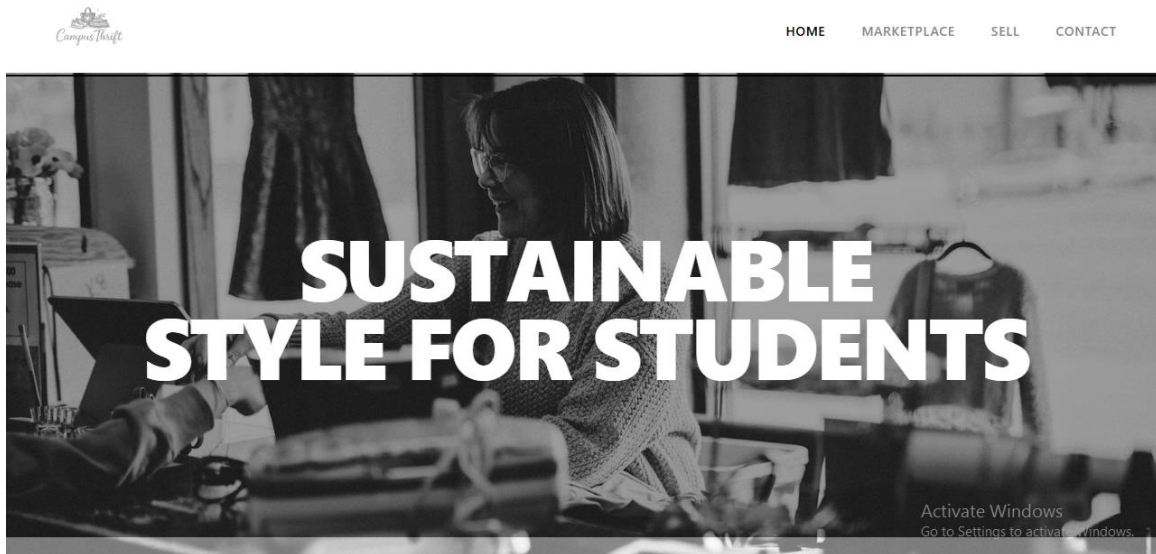
Animations and Transitions:

CSS transitions and hover effects are applied to buttons, product cards, images, and navigation links to enhance interactivity. Subtle animations improve user engagement while maintaining a professional appearance.

Sample Output Screenshots:

The following screenshots demonstrate the working of the Campus Thrift Store application:

1. Home Page showing hero section, categories, and featured items:



Shop by Category



CLOTHING



ACCESSORIES



SHOES

Activate Windows
Go to Settings to activate Windows.

Featured Products



LEATHER BAG

Rs 545

ACCESSORIES

Vintage brown leather messenger bag.

BUY NOW



VINTAGE WATCH

Rs 755

ACCESSORIES

Analog vintage style watch with leather strap.

BUY NOW



SNEAKERS

Rs 538

SHOES

Minimalist white sneakers, very comfortable.

BUY NOW

Activate Windows
Go to Settings to activate Windows.

2. Marketplace Page displaying dynamically loaded product cards:

[HOME](#)
[MARKETPLACE](#)
[SELL](#)
[CONTACT](#)

BACKPACK

Rs 118

ACCESSORIES

Durable waterproof backpack with laptop sleeve.

BUY NOW

OVERSIZED HOODIE

Rs 430

CLOTHING

Super soft oversized grey hoodie.

BUY NOW

VINTAGE DENIM JACKET

Rs 1200

CLOTHING

Classic 90s fit, light wash. No tears or stains, fits like a Large.

BUY NOW

Activate Windows
Go to Settings to activate Windows.



OVERSIZED FOOTBALL FC BARCELONA...

Rs 750

CLOTHING

Pre-loved heavyweight cotton FC Barcelona Pedit jersey. Perfectly baggy fit.

BUY NOW



SCIENTIFIC CALCULATOR

Rs 1000

ELECTRONICS

Casio-FX series, gently used for only one semester. All buttons responsive and batteries included.

BUY NOW



LEATHER CHELSEA BOOTS

Rs 2100

SHOES

Genuine espresso leather with elastic side panels. Features a nice worn-in patina, soles in great shape.

BUY NOW

Activate Windows
Go to Settings to activate Windows.



SILK PATTERNED SCARF

Rs 400

ACCESSORIES

Vibrant floral print, no snags or stains.

BUY NOW



WIRED RETRO HEADPHONES SET

Rs 550

ELECTRONICS

Foldable silver design, fully functional with 3.5mm jack.

BUY NOW



HANDMADE BRACELET

Rs 200

ACCESSORIES

One-of-a-kind piece featuring crystals and durable elastic cord.

BUY NOW

Activate Windows
Go to Settings to activate Windows.

3. Sell Page with item listing form and validation:

LIST YOUR ITEM

ITEM NAME

PRINTED KURTI

CATEGORY

CLOTHING

PRICE (RS)

800

DESCRIPTION

MINIMALIST PRINTED PINK KURTI WITH MEDIUM LENGTH AND FLORAL DESIGN, EXCELLENT CONDITION.

IMAGE

Choose File No file chosen

Please select a file.

ADD ITEM

Act
Go 1

4. Contact Page with submission alert:

CONTACT US

IRAJ

IRAJTARIQ84@GMAIL.COM

YOUR MESSAGE

Please fill out this field.

SEND

Activate Windows

